

Aditya Kumar Jha

GIS Engineer, Survey of India · Consulting Head, Uday GeoAnalytics Consultancy

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SUMMARY

Geoinformatics engineer at Survey of India's Geodetic & Research Branch. I process GNSS data from 1023+ CORS reference stations and levelling data for India's national vertical datum, and I write the Python software that automates this work. Before that, I worked on LiDAR, hyperspectral data and flood risk mapping at DRDO. I also run product at Uday GeoAnalytics Consultancy, where we build geospatial tools such as SolarSight. I have taken many projects from idea to finished product on my own, building them with AI-assisted development (vibe coding).

EXPERIENCE

GIS Engineer, Survey of India · Geodetic & Research Branch (G&RB) Jun 2025 – Present · Dehradun
HPL Wing (High Precision Levelling)

- Built a levelling network adjustment software for the HPL Wing that processes 1,00,000+ unique observations together, in real time. Earlier software could not adjust all observations at once. Computation for HPL work is now about 100x faster. This can be verified with the HPL Wing, Survey of India.
- Processed and validated levelling data for 1,00,780 Ground Control Points across India: height datum computation, loop closure analysis and systematic error correction.
- Automated geopotential number, standard deviation and Helmert orthometric height calculations, with QA/QC (Tau test, global statistical validation) built in.

NSRF Wing (GNSS CORS network)

- Wrote Python pipelines for GNSS data download, preprocessing and 365-day batch processing, which cut processing from 7 days to 3 days per station and save 12 hours of manual work every week.
- Processed multi-day GNSS data with GAMIT/GLOBK and PRIDE PPP-AR to centimetre-level station coordinates, in static and kinematic modes.
- Checked every observation file for cycle slips, data completeness and satellite geometry before processing.
- Worked with a team of 5 geodetic engineers to standardise data collection across 1023+ reference stations.

Consulting Head, Uday GeoAnalytics Consultancy 2023 – Present · India
Geospatial startup, formerly Urban Geo Dynamics. I lead an 8-member team across product, QA, content and strategy.

- SolarSight** (solarsight.in): free rooftop solar assessment for any address in India. Gives panel count, savings, subsidy, payback and installer leads as a 7-page PDF report in under 2 minutes, across all 36 states and union territories.
- Baadh Mitra**: flood early warning system for villages, with alerts on WhatsApp. In development.
- Jameen Mitra** (land records explained on WhatsApp), **Kuaan Mitra** (borewell success predictor) and **UGD LULC Engine** (land use classification for India). In development.

Research Intern, DRDO · Defence Geoinformatics Research Establishment (DGRE) Jul 2024 – Dec 2024 · Chandigarh

- Processed about 3 million LiDAR points and 32 hyperspectral bands in Python, using laspy and Spectral Python.
- Built flood inundation models in QGIS to find areas vulnerable to sea level rise of 1 m and above, combining 5+ layers (terrain, elevation, drainage) for defence planning.

SKILLS

Programming	Python (ArcPy, Pandas, NumPy, GeoPandas, Rasterio, laspy, Spectral Python, SciPy, GDAL/OGR, FastAPI), PyTorch, Next.js
GNSS & geodesy	GAMIT/GLOBK, PRIDE PPP-AR, RINEX, levelling network adjustment, Helmert orthometric heights
GIS & remote sensing	ArcGIS Pro, QGIS, Google Earth Engine, ENVI, ERDAS Imagine, Pix4Dmapper, GeoServer, MATLAB, AutoCAD (DWG/DXF)
Data	PostgreSQL/PostGIS, GeoJSON, Shapefile, Geodatabase, KML/KMZ, LiDAR point clouds, hyperspectral imagery

EDUCATION

B.Tech, Geoinformatics, Netaji Subhas University of Technology (NSUT), Delhi 2021 – 2025

- Founded the Geospatial Technology Club at NSUT and grew it to 200+ members from across India.

CERTIFICATIONS

- ArcGIS Pro Specialist (Esri Training) · Spatial Data Analysis with Google Earth Engine (CEPT University) · Drones and Drone Data Processing (CoELAM, Mysuru) · RS & GIS Application in Natural Resource Management (IIRS, Dehradun)